

GENERAL:

- 1) BUILDING IS TWO STORY.
- 2) FURNISH ALL LABOR, MATERIALS, TOOLS EQUIPMENT, FEES, PERMITS, CERTIFICATE OF INSPECTION, ETC, NECESSARY OR REASONABLY REQUIRED FOR THE COMPLETE INSTALLATION OF ALL AIR CONDITIONING WORK, THE WORK SHALL BE IN STRICT ACCORDANCE WITH THE ASHRAE GUIDE, AND ALL LOCAL AND STATE CODES, ORDINANCES AND REGULATIONS.
- 3) COORDINATE ALL MECHANICAL WORK WITH ARCHITECT AND OTHER TRADES PRIOR TO WORK.
- 4) ALL OUTSIDE AIR INTAKES SHALL BE MINIMUM 10 FEET FROM ANY EXHAUST OR PLUMBING VENTS - COORDINATE WITH PLUMBING CONTRACTOR.
- 5) EXHAUST DUCTS MUST TERMINATE 10 FEET HORIZONTALLY FROM OR 3 FEET ABOVE ALL AIR INTAKES.

EQUIPMENT:

- 1) MECHANICAL CONTRACTOR TO VERIFY AND COORDINATE AVAILABLE VOLTAGE WITH ELECTRICAL CONTRACTOR PRIOR TO ORDERING EQUIPMENT.
- 2) ALL MECHANICAL EQUIPMENT LOCATION TO COMPLY WITH THE 2010 CMC
- 3) SET A/C UNITS TO OUTSIDE AIR QUANTITIES AS SPECIFIED ON FLOOR PLAN.
- 4) ALL HVAC EQUIPMENT (A/C UNITS, EXHAUST FANS, ETC.) SHALL BE U.L. LISTED.
- 5) ALL MOTORS SHALL BE INHERENTLY THERMALLY PROTECTED.
- 6) CONTRACTOR SHALL MOUNT ALL MECHANICAL EQUIPMENT LEVEL IN CONFORMANCE WITH MANUFACTURER'S INSTRUCTIONS.
- 7) PROVIDE ONE SET OF FARR 30-30 THROWAWAY FILTERS AND ONE SET OF FINAL FILTERS, FRAMES, MOUNTING HARDWARE AND ACCESSORIES, PROVIDE ONE SET OF THROWAWAY FILTERS FOR USE DURING THE CONSTRUCTION PERIOD. REMOVE THE FIRST SET OF THROWAWAY FILTERS AFTER THE CONSTRUCTION PERIOD AND INSTALL THE SECOND SET OF FINAL FILTERS (PRIOR TO BALANCING THE SYSTEM),

DUCT WORKS:

- 1) THE MAXIMUM LENGTH OF ANY FLEX DUCT SHALL NOT EXCEED 8 FEET.
- 2) ALL SUPPLY AND RETURN DUCTS SHALL HAVE A MINIMUM OF R-5 INSULATION IN ACCORDANCE WITH IECC 803.2.8.
- 3) ALL JOINTS, LONGITUDINAL AND TRANSVERSE SEAMS AND CONNECTIONS IN DUCTWORK, SHALL BE SECURELY FASTENED AND SEALED WITH WELDS, GASKETS, MASTICS (ADHESIVES), MASTIC/PLUG-EMBEDDED FABRIC SYSTEMS OR TAPES. TAPES AND MASTICS USED TO SEAL DUCTWORK SHALL BE LISTED AND LABELED IN ACCORDANCE WITH UL 181A AND SHALL BE MARKED 181A-P FOR PRESSURE SENSITIVE TAPE. TAPES AND MASTICS USED TO SEAL FLEXIBLE AIR DUCT AND FLEXIBLE CONNECTORS SHALL COMPLY WITH UL 181B AND SHALL BE MARKED 181B-FX FOR PRESSURE SENSITIVE TAPE OR 181M FOR MASTIC. DUCT CONNECTION TO FLANGES OF AIR DISTRIBUTION SYSTEM EQUIPMENT SHALL BE SEALED AND MECHANICALLY FASTENED. UNLISTED DUCT TAPE IS NOT PERMITTED AS SEALANT ON ANY METAL DUCTS.
- 4) ALL DUCTS SHALL BE SUSTAINIALLY SUPPORTED WITH HANGERS TO THE STRUCTURE SHEET METAL SHALL BE IN ACCORDANCE WITH THE FOLLOWING SCHEDULE:  
UP TO 12' WIDTH 26 GAUGE STEEL  
13' TO 30' WIDTH 24 GAUGE STEEL
- 5) ALL "FACTORY MADE" DUCT MUST BE CLASS '0' OR CLASS '1'.
- 6) ALTERNATE FOR SHEET METAL DUCTWORK, FOR AIR CONDITIONING SLIPPLY AND RETURN: 1-1/2" THICK FIBERGLASS DUCTBOARD, WITH FOIL-SCRIM-DRAFT FIRE RESISTANT OUTER JACKET, MATERIAL SHALL COMPLY WITH UL-181, NFPA 90A AND 90B. CLOSURES SHALL BE MADE WITH THERMAL-ACTIVATED TAPE, OR CLOTH TAPE SET IN BRUSH APPLIED ADHESIVE, DO NOT UTILIZE PRESSURE SENSITIVE TAPE, DUCT BOARD SHALL BE EQUAL TO JOHNSMANSVILLE "MICRO-AIRE" M/F TYPE 475. TAPE SHALL BE EQUAL TO J. M. THERM-LOK-R OR PYRO-CLIRE 685.

OUTLETS:

- 1) THE MECHANICAL CONTRACTOR SHALL COORDINATE EXACT DIFFUSER AND GRILLE LOCATIONS WITH ELECTRICAL CONTRACTOR AND ALL OTHER TRADES AND ALSO COORDINATE SPACE AVAILABILITY FOR DUCTWORK ABOVE RECESSED LIGHTING TO AVOID RELOCATING DUCTWORK AT THE MECHANICAL CONTRACTORS EXPENSE.
- 2) ALL AIR DISTRIBUTION DEVICES IN LAY-IN CEILINGS SHALL BE SUPPORTED IN COMPLIANCE WITH SECTION 2, 3, 1, 3 OF NFPA 90A & 2007 CBC

CONTROLS

- 1) THERMOSTATS SHALL BE 7-DAY PROGRAMMABLE SET WITH FAN ON CONTINUOUSLY CARRIER MODEL #33CS DEBONAIR-FLATSTAT.

FIRE:

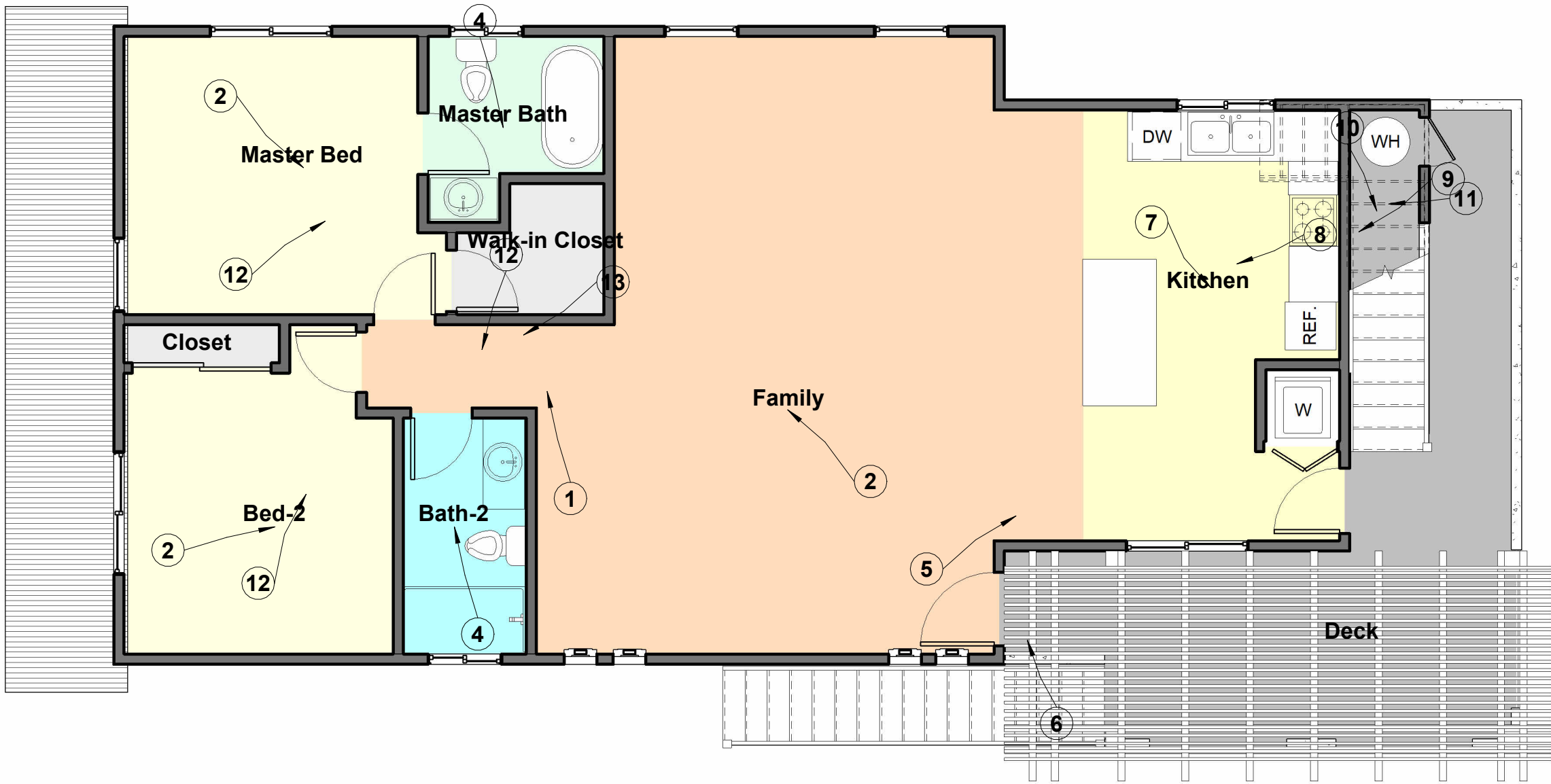
- 1) APPROVED SMOKE DETECTOR DEVICES SHALL BE INSTALLED WITH ALL AIR MOVING SYSTEMS \*EITHER INDEPENDENT OR COMBINED IN A COMMON PLENUM OR SERVING A COMMON ROOM) OVER 2000 CFM, LOCATE IN THE SUPPLY AIR DUCTWORK. DEVICES SHALL AUTOMATICALLY SHUT DOWN UNIT WHEN PARTICLES OF COMBUSTION ARE DETECTED WITHIN AIR STREAM. PER 2007 CMC.
- 2) ALL SMOKE DETECTORS SHALL BE TESTED BY AN AABC OR NEBB CERTIFIED AIR BALANCE TECHNICIAN PER THE CITY OF PHOENIX DEVELOPMENT SERVICES DEPARTMENT REQUIREMENTS FOR SPECIAL INSPECTIONS IN COMPLIANCE WITH THE 2007 C.M.C.. THE MECHANICAL CONTRACTOR SHALL INCLUDE ALL FEES FOR THESE TESTS AND INSPECTIONS IN HIS BID.
- 3) IF THERE IS FIRE ALARM SYSTEM, ALL SMOKE DETECTORS SHALL BE MONITORED BY THAT SYSTEM.

PIPING:

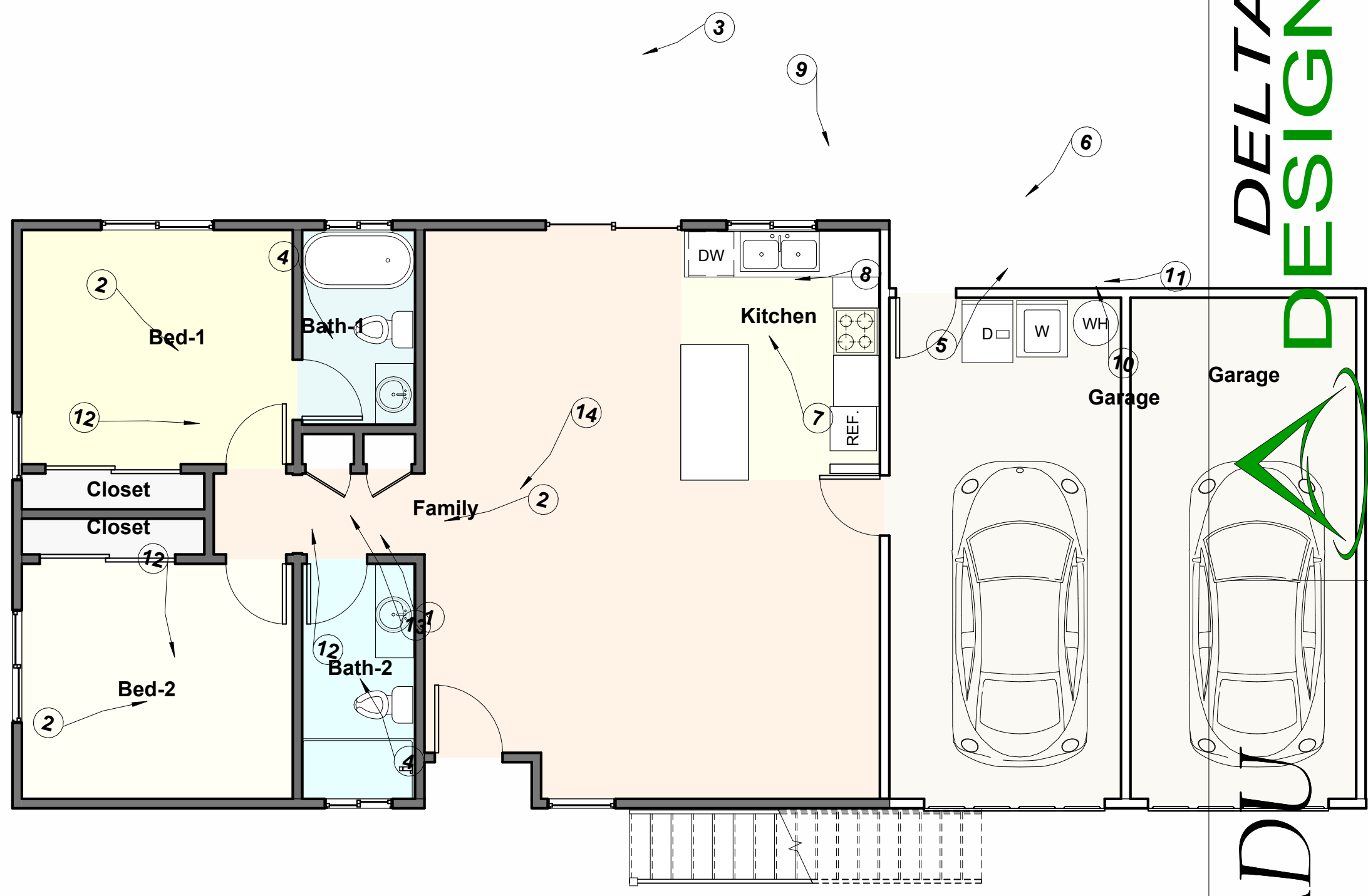
- 1) HANGERS AND SUPPORTS: CLEVIS HANGERS ADJUSTABLE HEIGHT, MAXIMUM 10' C, 3/8 "ROD UP TO 5" PIPES, 5/8" ROD FOR 6" AND LARGER. WHERE EXTRA HANGER SUPPORTS ARE REQUIRED, THEY SHALL BE PROVIDED BY CONTRACTOR, TRAPEZE HANGERS WITH ROLLERS MAY BE USED FOR MULTIPLE LINES OR WHERE SPACE IS LIMITED.

INSULATION:

- 1) ALL INSULATION, MATERIAL, COVERINGS, ADHESIVES, VAPOR-BARRIERS AND TAPES SHALL CONFORM TO NFPA 90A, FLAME SPREAD CLASSIFICATION NOT TO EXCEED 25 AND SMOKE: DEVELOPMENT NOT TO EXCEED 50.
- 2) PROVIDE 1/2" LINER IN SUPPLY AND RETURN DUCTWORK OF AIR CONDITIONING SYSTEMS
- 3) INSULATE ALL SHEET METAL DUCTWORK FOR AIR CONDITIONING SYSTEM WITH 1-1/2", 3/4 LB. DENSITY EXTERIOR INSULATION, WITH FSK JACKET AND VAPOR BARRIER.
- 4) INSULATION: R-5 SUPPLY AND RETURN AIR DUCT INSULATION IN UNCONDITIONED SPACES  
R-8 SUPPLY AND RETURN AIR DUCT INSULATION OUTSIDE THE BUILDING, R-8 INSULATION BETWEEN DUCTS AND THE BUILDING EXTERIOR WHEN DUCTS ARE PART OF A BUILDING ASSEMBLY  
EXCEPTION: DUCTS LOCATED WITHIN EQUIPMENT, EXCEPTION: DUCTS WITH INTERIOR AND EXTERIOR TEMPERATURE DIFFERENCE NOT EXCEEDING 15 DEG. F.



2 2nd Floor Mechanical Plan  
3/16" = 1'-0"



1 1st Floor Mechanical Plan  
3/16" = 1'-0"

MECHANICAL KEY NOTES:

- 1

THERMOSTAT / CONTROLLER MOUNTED AT 5'-0"
- 2

DUCTLESS MINI-SPLIT INDOOR UNIT (WALL MTD)
- 3

MINI-SPLIT CONDENSING UNIT ON CONC. PAD
- 4

CEILING EXHAUST FAN (MIN. OF 75CFM) MIN. OF 4" DIA. DUCTED TO OUTSIDE
- 5

4" DIA. DRYER EXHAUST DUCT TO OUTSIDE TOTAL OF TWO 90 DEGREE ELBOWS MAXIMUM 14' OF LENGTH
- 6

DRYER EXHAUST DUCT TERMINATION LOCATION
- 7

KITCHEN HOOD EXHAUST FAN (SEE SCHEDULE) WHICH MIN. OF 250CFM FOR INTERMITTENT
- 8

6" DIA. KITCHEN EXHAUST DUCT TO OUTSIDE BLDG
- 9

KITCHEN EXHAUST DUCT TERMINATION LOCATION (PROVIDE W/ RAIN CAP)
- 10

WATER HEATER P&T LINE TO OUTSIDE BLDG
- 11

HEAT PUMP WATER HEATER PER SCHEDULE
- 12

SMOKE DETECTOR, HARD-WIRED W/ BATTERY BACK-UP, INTERCONNECTED [CRC R314]
- 13

CARBON MONOXIDE ALARM [CRC R315]
- 14

IAQ (WHOLE BLDG VENT) EXHAUST FAN 64 CFM MIN. CONTINUOUS
- 15

| DRYER SCHEDULE |               |              |           |          |           |           |             |
|----------------|---------------|--------------|-----------|----------|-----------|-----------|-------------|
| MARK           | DESCRIPTION   | MANUFACTURER | Model     | LOCATION | DUCT SIZE | FUEL TYPE | DUCT TYPE   |
| DRYER 1        | CLOTHES DRYER | LG           | DLGX3471W |          | 4" DIA.   | GAS       | RIGID METAL |

| Exhaust Fan Schedule |                          |           |             |          |                |      |            |              |           |
|----------------------|--------------------------|-----------|-------------|----------|----------------|------|------------|--------------|-----------|
| MARK                 | DESCRIPTION              | LOCATION  | MANUFACTURE | MODEL    | AIR FLOW (CFM) | SONE | DUCT SIZE  | WEIGHT (LBS) | NOTES     |
| EF1                  | Ceiling Exhaust Fan      | Bath room | Broan       | QTRE100S | 110            | 1.5  | 4"         | 12.6         |           |
| EF2                  | Kitchen Hood Exhaust Fan | Kitchen   | Broan       | E6436ss  | 600            | 1.5  | 3-1/2"x10" |              | UL#E17814 |

| WATER HEATER SCHEDULE (HEATPUMP ) |             |             |          |     |                |             |          |           |          |             |              |               |
|-----------------------------------|-------------|-------------|----------|-----|----------------|-------------|----------|-----------|----------|-------------|--------------|---------------|
| DESCRIPTION                       | MANUFACTURE | MODEL       | CAPACITY | KW  | Supply Fitting | Supply Pipe | VOLTAGE  | VENT TYPE | DIAMETER | HEIGHT (FT) | RECOVER RATE | ENERGY FACTOR |
| WATER HEATER                      | AO SMITH    | FPTU 50 120 | 50 GAL   | 4.5 | 3/4"           | 1"          | 208/240V | PVC       | 22"      | 5' - 3"     |              |               |



| Revision Dates |          |
|----------------|----------|
| 1              | 08.15.24 |
| 2              | 01.01.25 |

SHEET TITLE  
*Mechanical Plan*

Drawing Date 08.27.26

Check By  
Francis N.

Scale As indicated

Job Number 022225

Sheet Number

A200